

We observe two independent samples: prior- and current-scores, X and Y , respectively, for the same subgroup S (e.g., a TIMSS country or a NAEP state). We map scores to reference-quantiles using fixed reference marginals:

1. Prior reference quantiles $U = F_X^{\text{ref}}(x_i)$ and
2. Current reference quantiles $V = F_Y^{\text{ref}}(y_j)$.

Crucially, there is *no student-level linkage* between these samples—we cannot match any u_i to its corresponding v_j .

Panel A shows the two *univariate* marginals plotted on the same reference-quantile axis (“ u or v ”). This is the situation faced by TIMSS/NAEP-style designs when cohorts are sampled independently.

The distributions U and V are the only observables. All subsequent inference proceeds from these marginals plus an externally supplied copula dependence structure.